Training a Regression Model on Azure Platform

Objective:

Train a regression model using a dataset (random or manual dataset created by Maria) on the Azure Machine Learning platform to predict target values based on input features without doing any coding.

Create DataSet in Excel and Import as CSV File:

	A	B	C	D
1	Serial No.s	Names	Age	Salary
2	1	Ali	25	3928
3	2	Khan	44	1546
4	3	Sara	40	2397
5	4	Nick	48	1579
6	5	Johns	43	2323
7	6	Sana	39	3374
8	7	Aisha	24	1198
9	8	Taylor	21	4339
10	9	Jira	31	2029

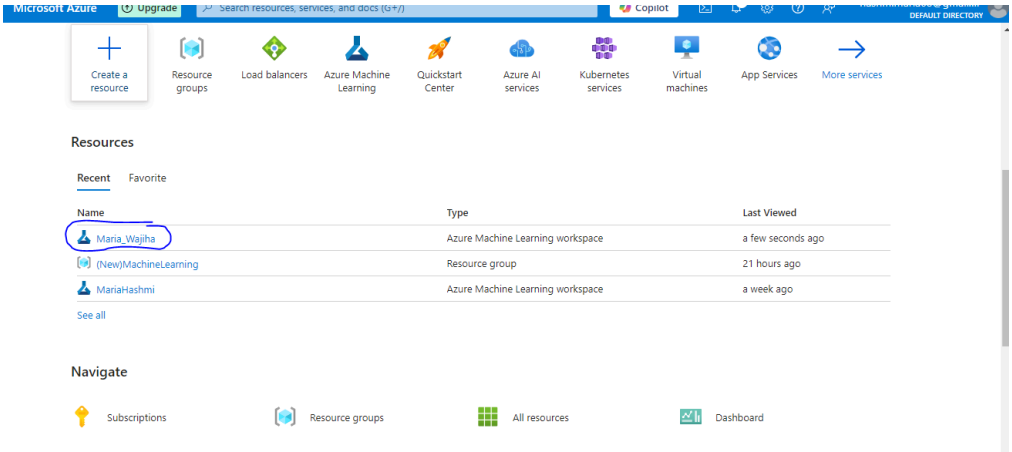
.....

A42 fx 41

	A	B	C	D
42	41	Jira	43	3006
43	42	Richa	15	1342
44	43	Roomi	31	2524
45	44	Ash	39	4599
46	45	Rock	33	3965
47	46	Max	50	3357
48	47	Tony	36	2813
49	48	Robin	49	2914
50	49	Ali	55	1809
51	50	khan	60	4647

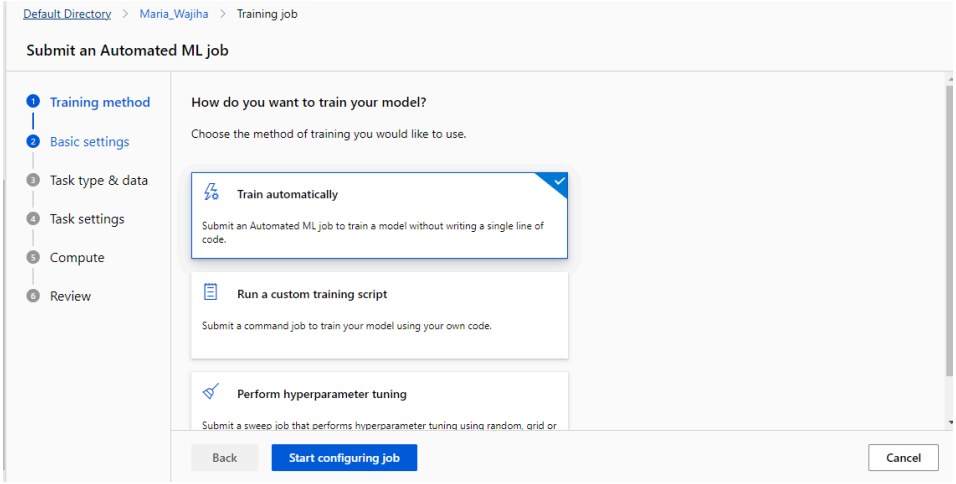
Step 1: Create an Azure Machine Learning Workspace

- Sign in to Azure portal and create a new resource.
- Search for Machine Learning via search bar and select Create.
- Fill in the required details like subscription, resource group, and workspace name, then Create the workspace (step by step).



Step 2: Prepare the Dataset

- Prepare the dataset in CSV file.
- Upload the file of dataset to the Azure Blob Storage after that follow NEXT step to proceed.
- Click on Datasets under the Assets tab, then Create Dataset and fill blank fields according to suitable name related to your work to understand easily.
- Then choose From local files.
- Define the dataset properties like name, description, and data type. Verify and register the dataset.



Default Directory > Maria_Wajiha > Training job

Submit an Automated ML job

- Training method
- Basic settings
- Task type & data
- Task settings
- Compute
- Review

Job name *

Maria

Experiment name *

Select existing | Create new

New experiment name *

Maria_Regression

Description

Training ML model without coding

Tags

Name : Value Add

Back | Next | Cancel

Default Directory > Maria_Wajiha > Training job

Submit an Automated ML job

- Training method
- Basic settings
- Task type & data
- Task settings
- Compute
- Review

Task type & data

Choose the type of task that you would like your model to perform and the data to use for training. Learn more

Select task type *

Regression

Select data

Make sure your data is preprocessed into a supported format.

+ Create | Refresh | Show supported data assets only | Reset view

Search | Filter | Columns

Name	Type	Created on ↓	Modified on

Back | Next | Cancel

Create data asset

- Data type
- Data source

Set the name and type for your data asset

Name *

Maria_Regression

Description

Data asset description

Type *

Tabular

Use cases for data types

When should I use File type?

The File type is recommended in most scenarios when you are working with a single data file of any type (including tabular data). This type allows you to specify a file location by URI in a storage location on your local computer, an attached Datastore, blob/ADLS storage, or a publicly available http(s) location. There are many types of supported URIs. In the Azure Machine Learning CLI v2 or Python SDK v2, this data type is called uri_file. Learn more about the uri_file type

When should I use Folder type?

The Folder type has all the same capabilities and use cases as the File type, but is used when specifying a

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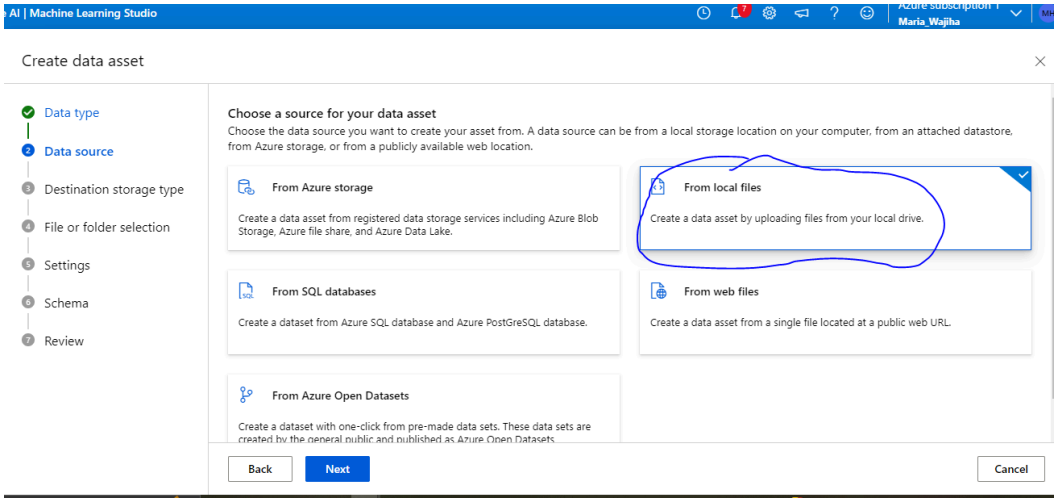
AI | Machine Learning Studio
Maria_Wajiha
Create data asset
Data type
Data source
Destination storage type
File or folder selection
Settings
Schema
Review
Choose a source for your data asset
Choose the data source you want to create your asset from. A data source can be from a local storage location on your computer, from an attached datastore, from Azure storage, or from a publicly available web location.
From Azure storage
Create a data asset from registered data storage services including Azure Blob Storage, Azure file share, and Azure Data Lake.
From local files
Create a data asset by uploading files from your local drive.
From SQL databases
Create a dataset from Azure SQL database and Azure PostGreSQL database.
From web files
Create a data asset from a single file located at a public web URL.
From Azure Open Datasets
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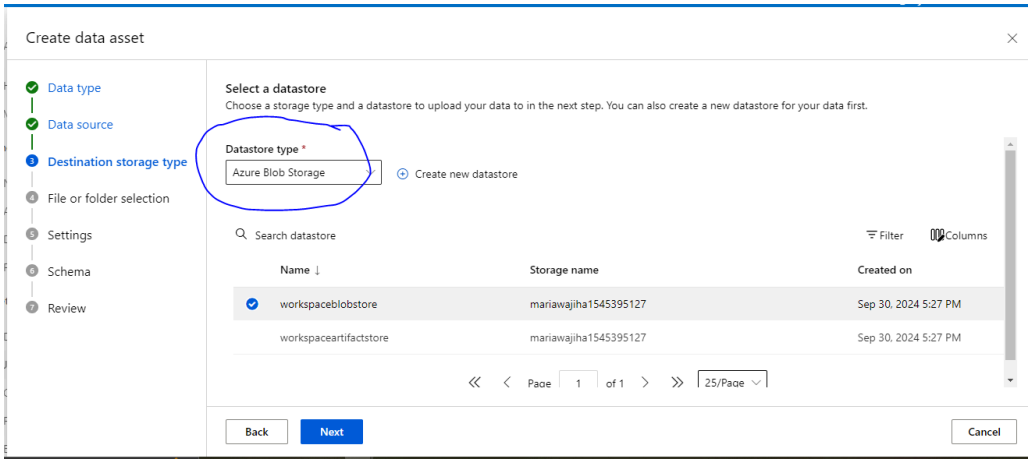
Cancel


Create data asset
Data type
Data source
Destination storage type
File or folder selection
Settings
Schema
Review
Select a datastore
Choose a storage type and a datastore to upload your data to in the next step. You can also create a new datastore for your data first.
Datastore type *
Azure Blob Storage
Create new datastore
Search datastore
Filter
Columns
Name
Storage name
Created on
workspaceblobstore
mariawajiha1545395127
Sep 30, 2024 5:27 PM
workspaceartifactstore
mariawajiha1545395127
Sep 30, 2024 5:27 PM
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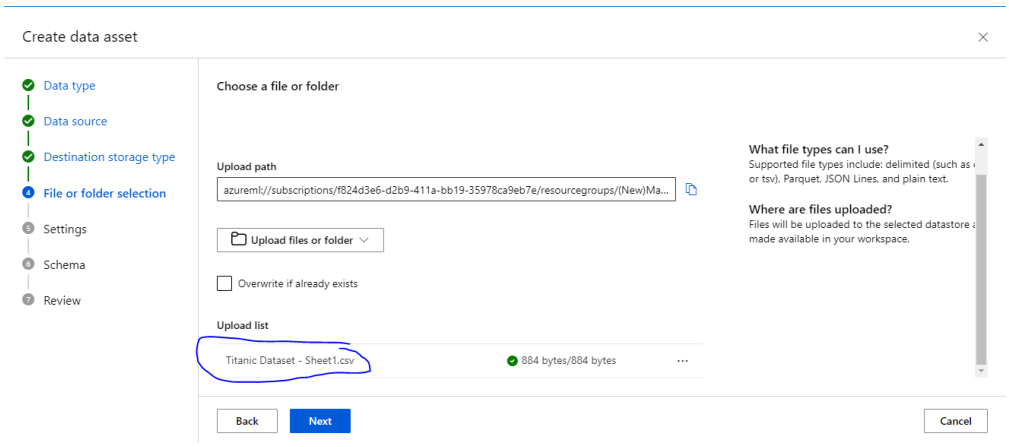
Cancel


Create data asset
Data type
Data source
Destination storage type
File or folder selection
Settings
Schema
Review
Choose a file or folder
Upload path
azureml://subscriptions/f824d3e6-d2b9-411a-bb19-35978ca9eb7e/resourcegroups/(New)Ma...
Upload files or folder
Overwrite if already exists
Upload list
Titanic Dataset - Sheet1.csv
884 bytes/884 bytes
What file types can I use?
Supported file types include: delimited (such as
or tsv), Parquet, JSON Lines, and plain text.
Where are files uploaded?
Files will be uploaded to the selected datastore
made available in your workspace.
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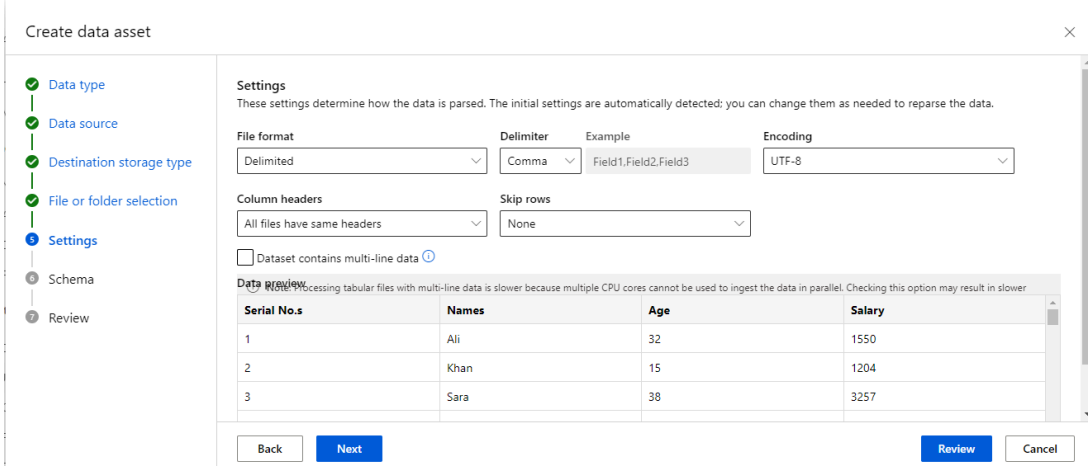
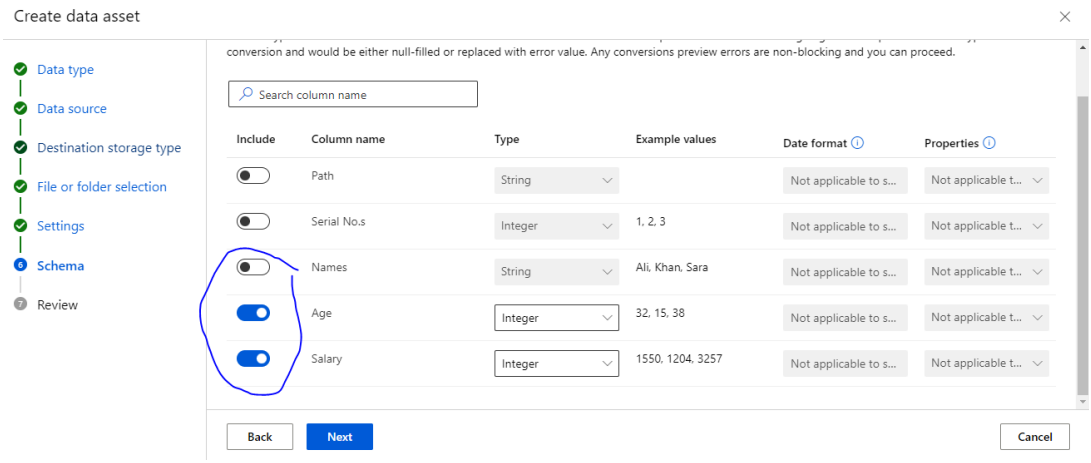
Cancel





Step 3: Create an Experiment

- In the Azure Machine Learning Studio, navigate to the Experiments section or portion.
- Click Create new and provide a name for the experiment. (we can assign any name to understand easily that, which model we applied)
- This experiment will serve as the container for tracking your model training and performance metrics.
- After completion your task you can can your check your summary between the matrices parameters.

Azure AI | Machine Learning Studio

Azure subscription 1
Maria_Wajiha

All workspaces

Home
Model catalog

Authoring

Notebooks
Automated ML
Designer
Prompt flow

Assets

Data
Jobs
Components
Pipelines
Environments

Default Directory > Maria_Wajiha > Jobs > Maria_ML > Maria_Dataset

Maria_Dataset

Completed

Overview | Data guardrails | Models + child jobs | Outputs + logs | Child jobs

Refresh | Edit and submit (preview) | Register model | Cancel | Delete | Compare (preview)

Properties

Status
Completed

Warning: Experiment timeout reached, hence experiment stopped. Current experiment timeout: 6 hour(s) 0 minute(s)

See more details

Created on
Oct 7, 2024 7:33 PM

Start time
Oct 7, 2024 7:33 PM

Duration

Inputs

Input name: training_data
Data asset: Maria_Titanic:1
Asset URI: azureml:Maria_Titanic:1

Outputs

Output name: best_model
Model: azureml_maria_dataset_352_output_mlflow_log_model_920577991:1
Asset URI: azureml:azureml_maria_dataset_352_output_mlflow_log_mode...

Output name: full_training_dataset
Dataset: 69de5cbf-b67e-4d39-ab58-6d352fb76893

Default Directory > Maria_Wajiha > Jobs > Maria_ML > Maria_Dataset

Maria_Dataset

Completed

Overview | Data guardrails | Models + child jobs | Outputs + logs | Child jobs

Duration
6h 8m 41.34s

Compute duration
6h 8m 41.34s

Name
maria_dataset

Script name
--

Created by
Maria Hashmi

Job type
Automated ML

Experiment
Maria_ML

Dataset: 69de5cbf-b67e-4d39-ab58-6d352fb76893

Best model summary

Algorithm name
VotingEnsemble

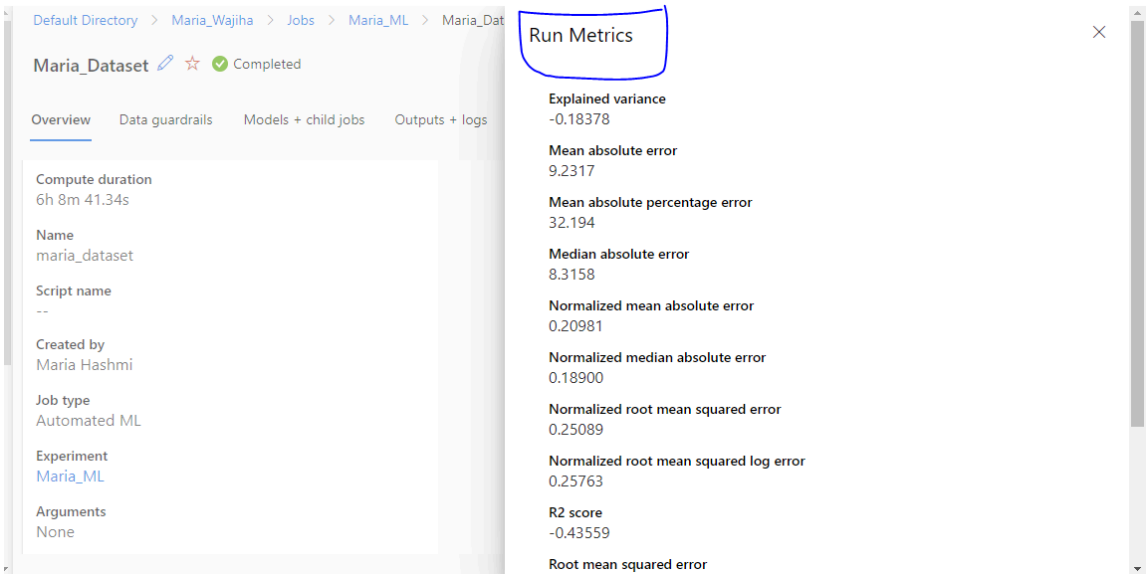
Ensemble details
View ensemble details

Normalized root mean squared error
0.25089 View all other metrics

Sampling
100.00 %

Registered models
No registration yet

Deploy status



Step 4: Plotting Graph

Here is the regression model graph showing the relationship between age and salary (just integer values) from the dataset. The blue points represent the actual data from my dataset, while the red line represents the linear regression line (applied).

Step 5: Conclusion

Using Azure's integrated environment, we can efficiently build and train a regression model. This process facilitates seamless scalability, real-time monitoring, and integration capabilities for predictive analytics without doing any coding.

Submitted By: Maria Hashmi